Gaussian Mixture Models, EM Algorithm, and Pattern Classification

Lab Assignment 7 Report

Shrawanthi P. (Roll No: 26120098)
Shreepriyadarshini RM. (Roll No: 26120099)

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1 Objective

The objective of this assignment are as follows:

1. To Build Gaussian Mixture Models (GMMs) for each class using the 2D synthetic dataset from Assignment 6 (trained on 80% of the data) and evaluate classification accuracy on the 20% test data.
2. To work with the handwritten digits assigned to our team (digits **1, 8, and 9**) from the MNIST dataset.
3. The images can be compressed from 28×28 pixels down to 14×14 pixels to reduce feature size while keeping the core stroke shapes.
4. And then train a diagonal-covariance GMM for each digit using the Expectation-Maximization (EM) algorithm on the training set only.
5. Monitor the model convergence across EM iterations on both the training set and the dev set.
6. Classify the test images using the learned models, compute the confusion matrix, and explain misclassifications by inspecting decision margins and class boundaries.

2 Workflow

The experiment is divided into two parts:

1. **Part 1: 2D Synthetic Data Classification**
   - **Data & Split:** We reuse the two-class synthetic dataset generated in the previous assignment 6 (4,000 points per class, 3 Gaussian components each). Each class is split into 80% training data (3,200 points) and 20% test data (800 points).
   - **GMM Training:** Using random initialization, we fit a GMM with full covariance for each class using the EM [Expectation Maximization] algorithm.
   - **Classification:** Test samples are classified by picking the class with the higher log-likelihood ($\log p(\mathbf{x} \mid c)$).
2. **Part 2: MNIST Handwritten Digit Classification (Team 3: Digits 1, 8, 9)**
   - **Dataset Preparation:** We extract the images from the dataset. The dataset contains separate folders for training (14,833 images), development (3,709 images), and test (3,118 images).
   - **Image Compression:** Every 28×28 image is downsampled or compressed to 14×14 pixels ($D = 196$ features), normalized to values between 0 and 1. Original 28×28 images are preserved to inspect the misclassified examples.

- **Model Fitting using EM:** For each digit class, we train a GMM with $K = 3$ Gaussian components and diagonal covariance matrices. We have choosen 35 EM iterations for training dataset. A small variance floor (10^{-3}) prevents numerical errors on blank background pixels.
- **Evaluation:** We then monitor the average log-likelihood on both training and development sets at each EM step. The final predictions on the test set are obtained by choosing the digit that yields the largest log-likelihood.

3 Outputs and Results

3.1 Part 1: 2D Synthetic Data Results

Table 1 shows the confusion matrix for the 2D synthetic test points, and Figure 1 displays the decision boundaries side-by-side.

True Class	**Predicted Class 1**	**Predicted Class 2**
Class 1	800	0
Class 2	0	800

Table 1: Confusion matrix on synthetic 2D test data (Accuracy: 100.0%).

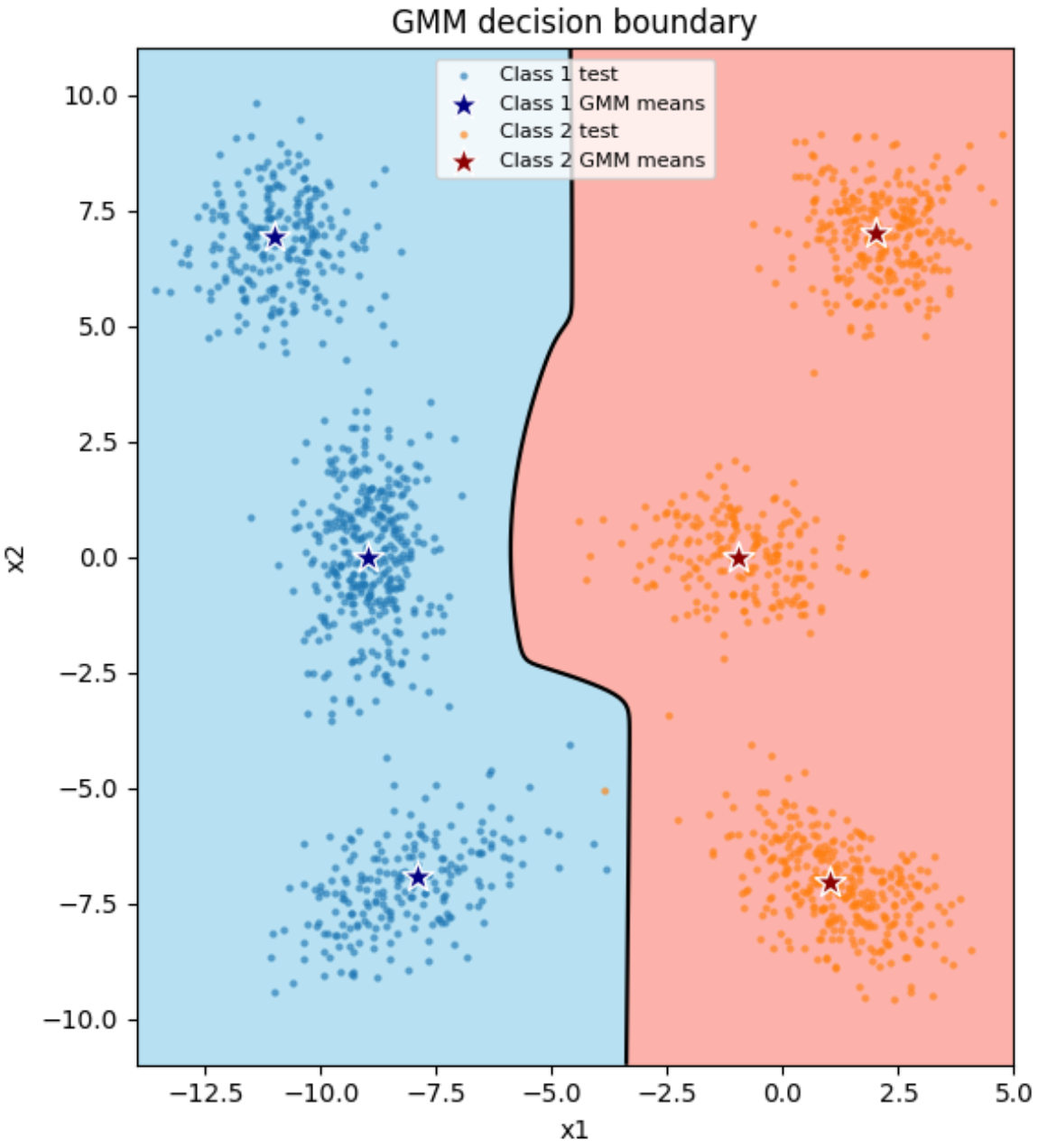


Figure 1: GMM decision boundaries (log-likelihood) .

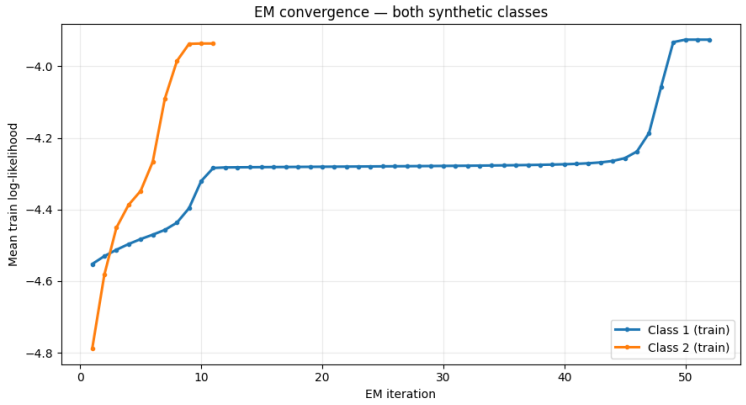


Figure 2: EM convergence curve for synthetic Class 1 and Class 2.

As shown in Figure 1, both methods separate the two classes cleanly because the classes are far apart. However, while K-Means only considers the distance to cluster centers, the GMM boundary accounts for the shape, orientation, and spread of the clusters. Figure 2 The EM convergence graph shows different learning behaviours for the two synthetic classes. Class 2 converges rapidly within approximately 10 iterations, with a steady increase in log-likelihood. Class 1 initially improves slowly and remains nearly constant for several iterations before showing a sharp increase around iteration 48, finally converging near iteration 50.

3.2 Part 2: MNIST Digit Image Representation

Figure 3 shows the original 28×28 digits along with their compressed 14×14 representations. Compressing the images reduces the feature dimension from 784 to 196 while still keeping the primary handwritten stroke structure clear.

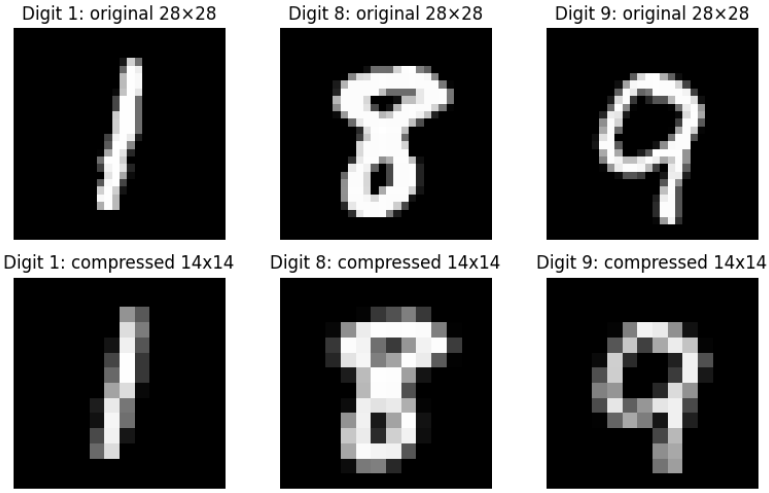


Figure 3: Original 28×28 handwritten digits (top) and compressed 14×14 features (bottom) for digits 1, 8, and 9.

3.3 Part 2: EM Convergence for MNIST Digits

Figure 4 shows the log-likelihood curves across 35 EM iterations for both the training set and the development set.

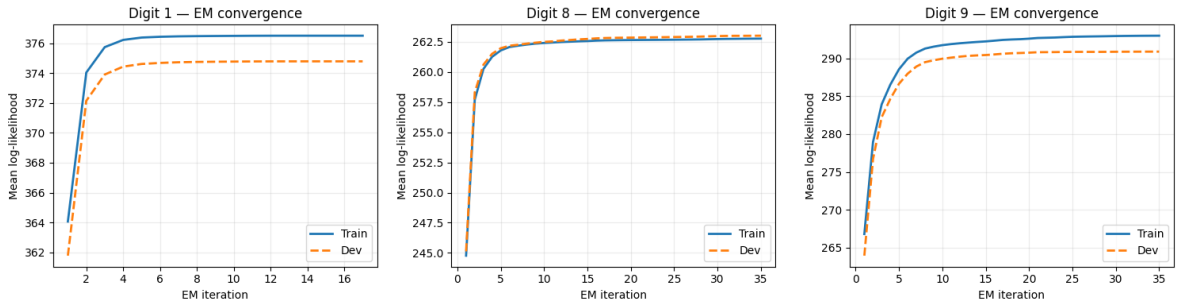


Figure 4: EM convergence curves for digits 1, 8, and 9 showing training and development log-likelihoods over 35 iterations.

The training log-likelihood increases gradually with every iteration as guaranteed by the EM formulation, showing that the model generalizes well without overfitting.Digit 1 converges quickly, stabilizing around iteration 6. Digit 8 improves gradually and stabilizes around iteration 25, with development likelihood slightly higher than training. Digit 9 rises sharply at first, then improves slowly before stabilizing around iteration 30.

3.4 Part 2: MNIST Classification and Confusion Matrix

Table 2 gives the dataset sizes, and Table 3 presents the confusion matrix on the 3,118 test images. Figure 5 shows the confusion matrix plot.

Digit	Training Samples	Development Samples	Test Samples	Total
Digit 1	5,383	1,359	1,135	7,877
Digit 8	4,674	1,177	974	6,825
Digit 9	4,776	1,173	1,009	6,958
Total	**14,833**	**3,709**	**3,118**	**21,660**

Table 2: Sample distribution across splits for Team 3 digits.

True Digit	Pred. 1	Pred. 8	Pred. 9	Recall
Digit 1	**1110**	24	1	97.80%
Digit 8	22	**926**	26	95.07%
Digit 9	8	50	**951**	94.25%

Table 3: Test set confusion matrix for digits 1, 8, and 9 (Overall Accuracy: **95.80%**).

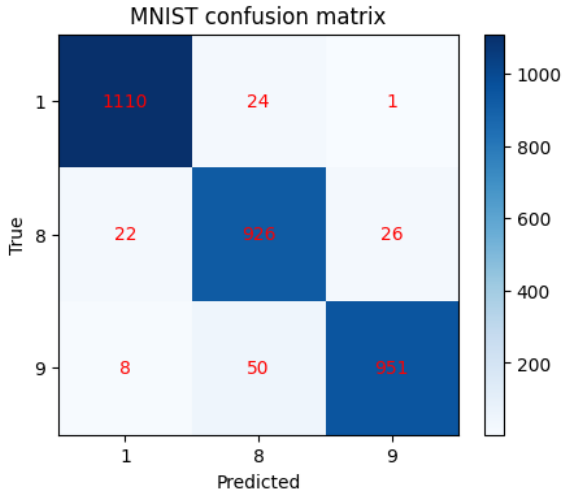


Figure 5: Heatmap of the test confusion matrix for digits 1, 8, and 9.

The trained GMM models achieve an overall test accuracy of **95.80%** (2,987 out of 3,118 test images classified correctly). Digits 8 and 9 are misclassified in few cases.

3.5 Part 2: Visualizing Learned GMM Parameters

Figure 6 shows the learned mean vectors $\boldsymbol{\mu}_k$ for each of the $K = 3$ Gaussian components reshaped to 14×14 pixels, along with their mixing weights π_k.

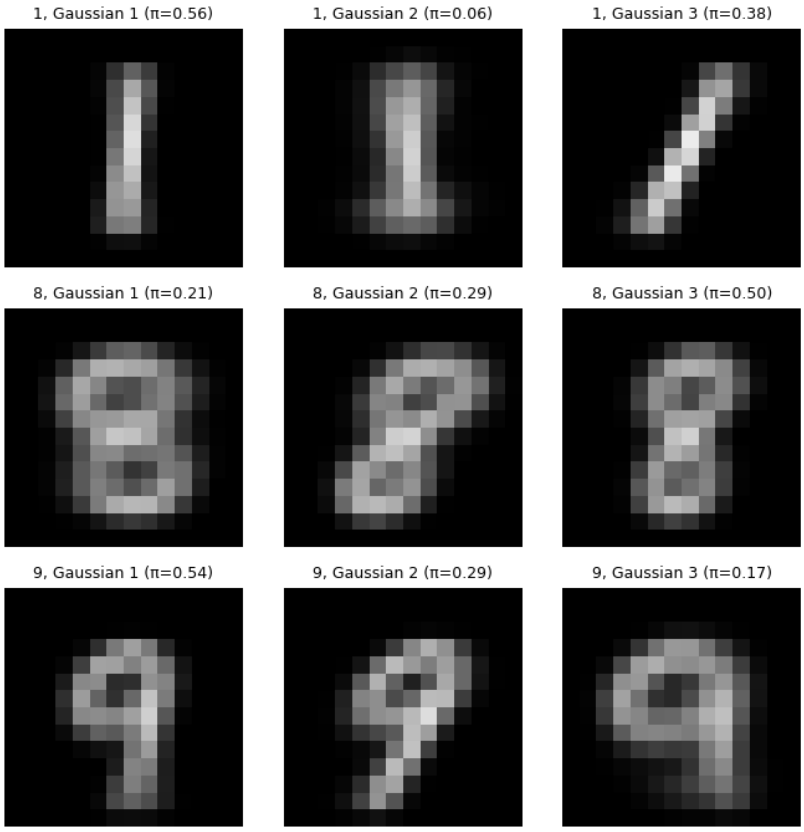


Figure 6: Learned component means (average digit templates) and mixing weights π for each digit model.

Each component captures a prominent handwriting style:

- **Digit 1:** It captures the straight vertical strokes, slightly slanted strokes, and strokes with a slight top hook.
- **Digit 8:** It captures symmetrical vertical eights, wider eights, and eights with different top and bottom loop size.
- **Digit 9:** It captures rounded top loops with a straight vertical stem, curved stems, and slanted stems.

3.6 Part 2: Error Analysis and Misclassified Samples

Figure 7 displays selected misclassified test images (in their original 28×28 resolution) that had the smallest decision margins (difference between the top two log-likelihood scores).

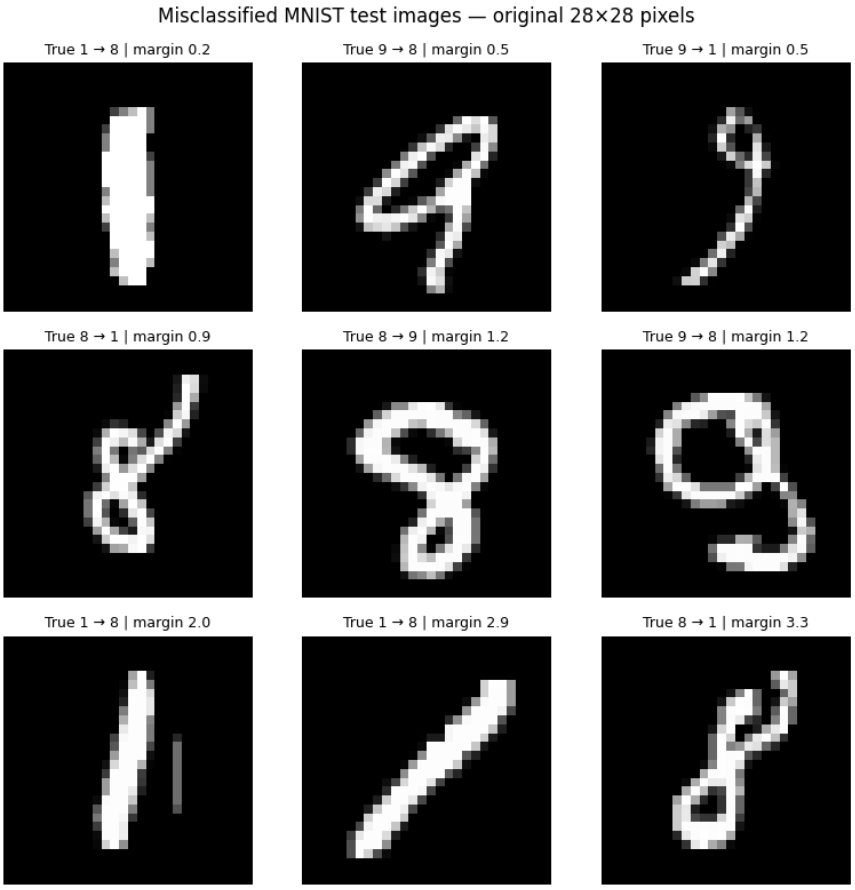


Figure 7: Examples of misclassified test digits with their true label, predicted label, and log-likelihood decision margin.

4 Findings

- **GMM versus K-Means Decision Boundaries:** On the 2D synthetic dataset, both classifiers achieve 100% test accuracy as there is a large separation between the classes. However, the GMM creates smooth curved boundaries because it considers the shape, spread and component weights, whereas K-Means creates straight-line boundaries based solely on distance.

- **EM Stability and Convergence:** The log-likelihood increased with each iteration and gradually stabilized. Convergence varied across datasets, with some models stabilizing within 10–30 iterations, while synthetic Class 1 took around 50 iterations. A small variance floor (10^{-3}) is used to prevent numerical underflow on empty background pixels without harming model performance.
- **Digit Confusions and Class Boundaries:** Out of 3,118 test images, only 131 were misclassified. Looking at the class boundaries and confusion patterns:
  - *Digit 9 confused with Digit 8 (50 cases) and 8 with 9 (26 cases):* This is the most frequent confusion. Both digits have a loop at the top, so some handwritten 8s and 9s look very similar. The two models produce nearly identical log-likelihood values (very small decision margins).
  - *Digit 1 confused with Digit 8 (24 cases) and 8 with 1 (22 cases):* Some narrow or quickly written 8s look like digit 1. Likewise, thick or slanted 1s can resemble part of an 8.
  - *Digit 1 and Digit 9 are rarely confused (only 1 case of* $1 \to 9$ *and 8 cases of* $9 \to 1$*):* These digits were rarely confused because digit 1 is usually a straight line, while digit 9 has a clear loop.
- **Effectiveness of Compression:** The images were reduced from 28×28 to 14×14 pixels which reduced the feature dimension by 75% (from 784 down to 196). This made GMM training faster and memory-efficient while still retaining a high classification accuracy of 95.80%.